CS 2490 – Summative Homework 2

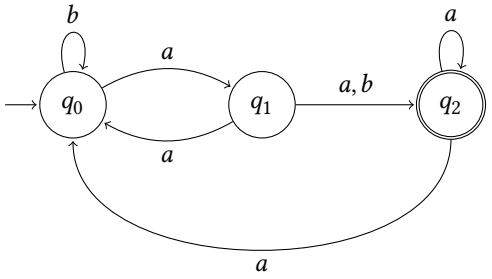
Turn in by 2026/10/31 at 11:59pm

You **must** turn in your assignment as a single zip file entitled: *lastname_firstname_shw2.zip*. It should contain a single pdf file named *lastname_firstname_shw2.pdf* containing your answers to the exercises below (either typed or handwritten and scanned). The zip file will also contain the JSON files required by each exercise that asks for them. Name them as follows: *lastname_firstname_shw2_exX.json* where X is the exercise number.

Exercise 1. Show that the language $L_1 = \{0^i 1^j \mid i > j \geq 0\}$, over the alphabet $\{0, 1\}$, is not regular.

Exercise 2. Let Σ be any alphabet with at least two letters. Show that the language $L_2 = \{w \in \Sigma^* \mid w \neq w^R\}$ is not regular.

Exercise 3. Use the state-removal algorithm seen in class to convert the following NFA into a regular expression. Make the intermediate steps apparent: draw the generalized NFA obtained after each state removal.



Exercise 4. Use the algorithm seen in class to convert the regular expression $(1(01 + 10)(0 + 1)^*)^*0$ into an NFA. Make all your intermediate automata explicit. Include the JSON file representing the final NFA you obtain.

Exercise 5. Consider the grammar $G = (\{a, b\}, \{S, A\}, S, P)$ where P consists of the following production rules:

$$S \to abS \mid aA, \qquad A \to bA \mid \varepsilon$$

(i) Use the construction seen in class to build an NFA $\mathscr{A}$ that recognizes $\mathrm{L}(G)$. Include the JSON file of $\mathscr{A}$ in your submission.

(ii) Apply the subset construction to $\mathscr{A}$.

(iii) Deduce a regular grammar that generates the complement $\{a, b\}^* \setminus \mathrm{L}(G)$.[1]

[1] Consistency will be taken into account: a correct treatment of an incorrect previous answer still gets full credit.